Sparse Gaussian Gradient Code

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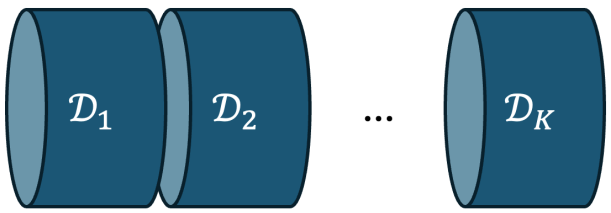
Problem Formulation

Distributed Computing

Goal: Speedup learning process on large-scale data sets.



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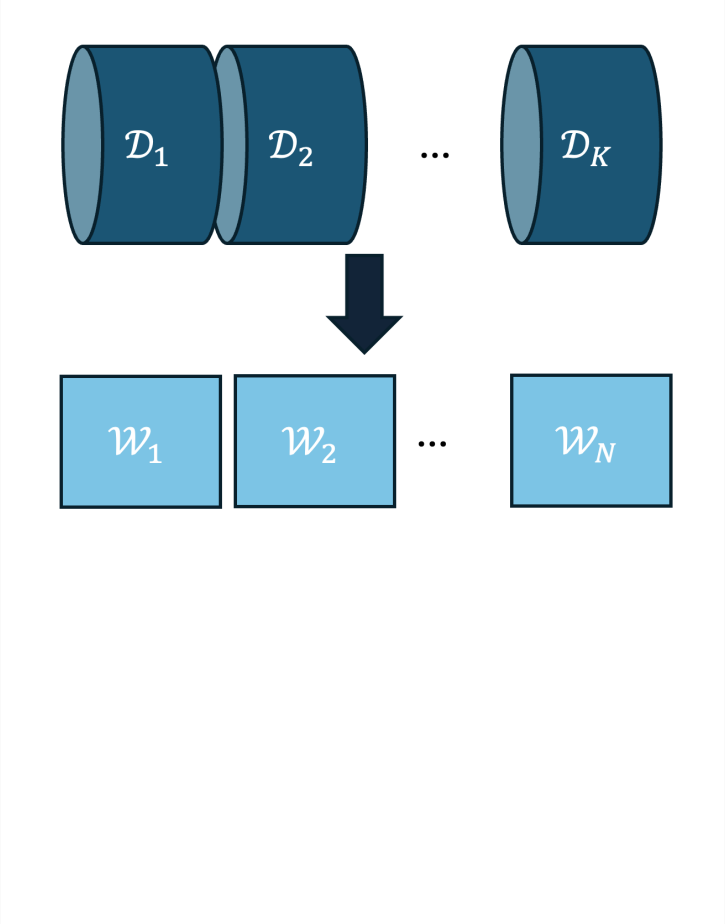

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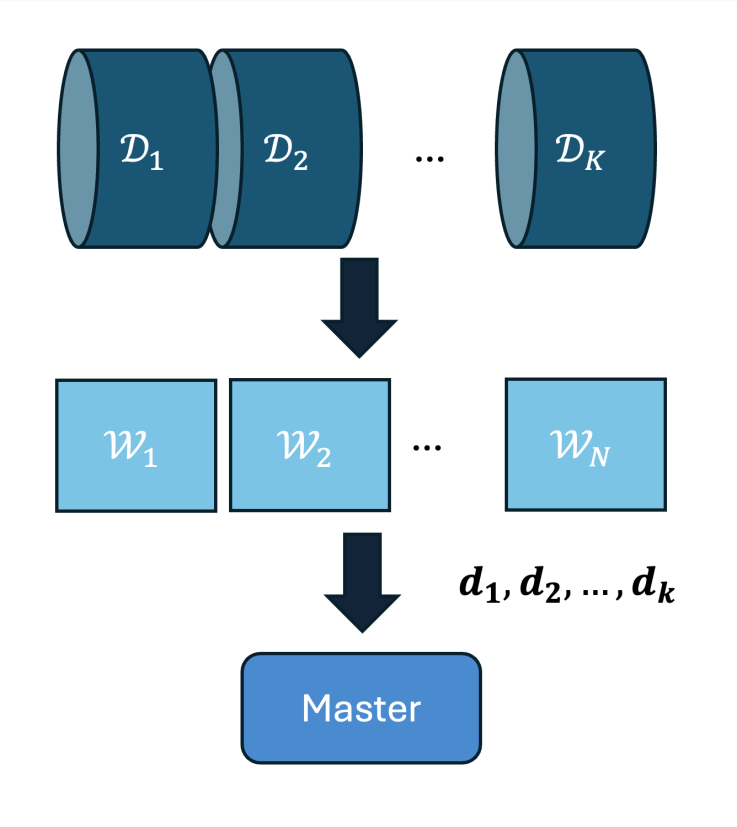

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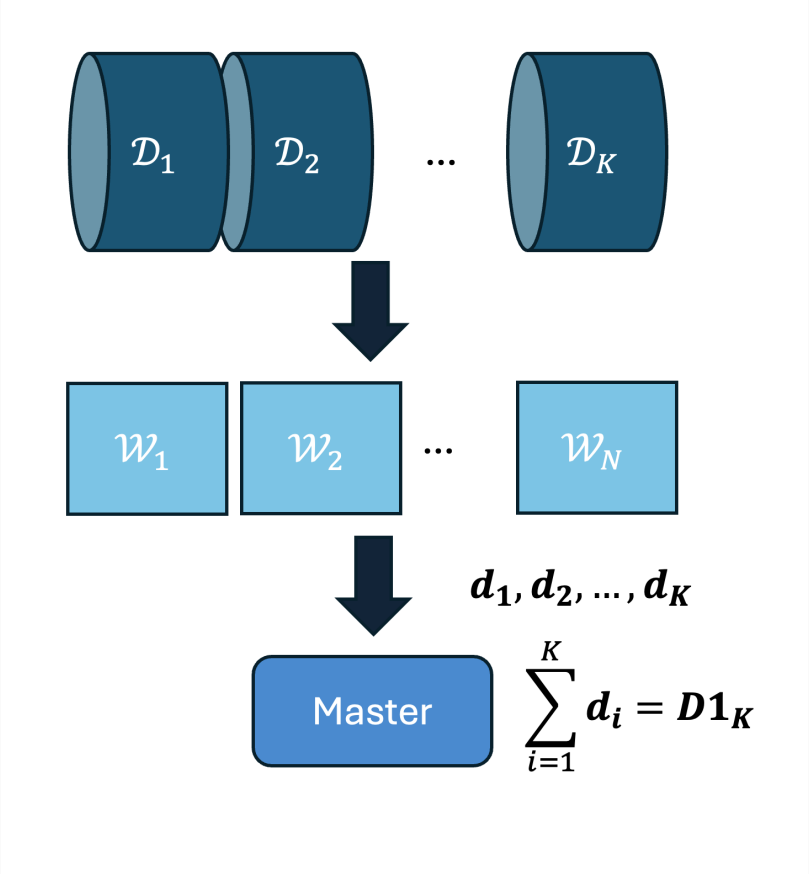

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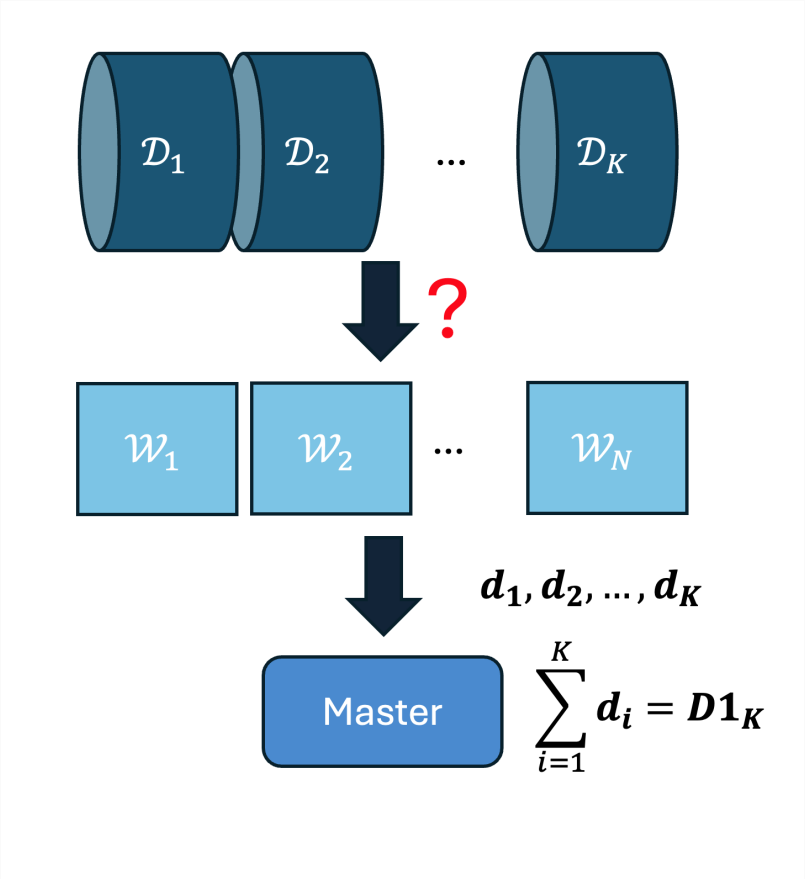

How to design the distribution of computing tasks?

Gradient Coding: Exact Recovery

Gradient Coding emerges as a tool to design the distribution of computing tasks [Tandon et al. (2017)].

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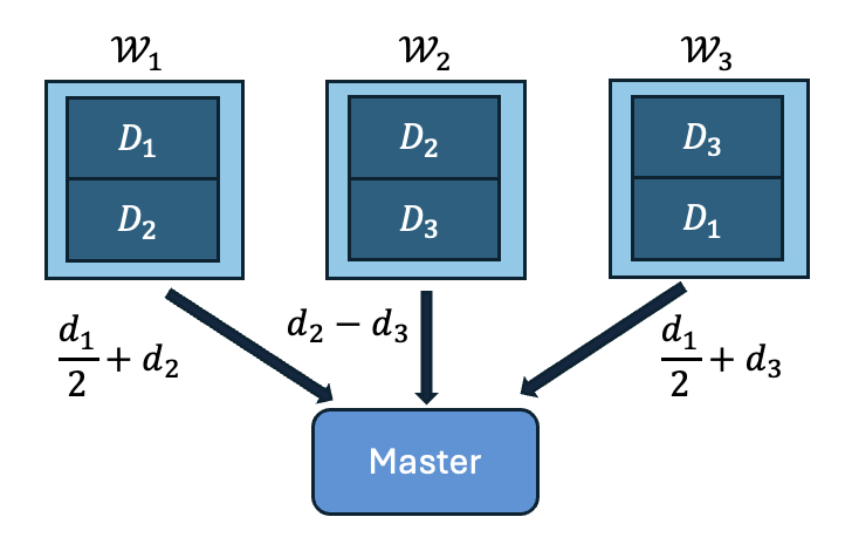

$$\boldsymbol{E} = \begin{pmatrix} 1/2 & 0 & 1/2 \\ 1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$$

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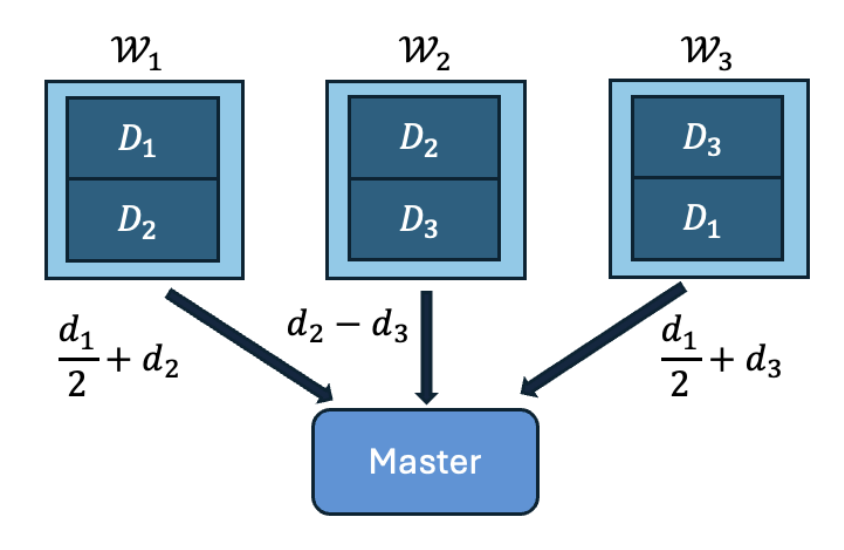


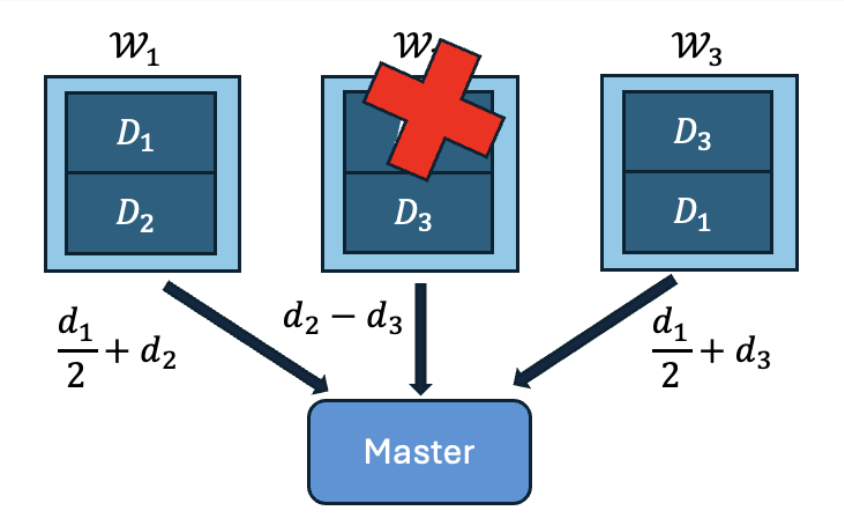
$$\boldsymbol{E} = \begin{pmatrix} 1/2 & 0 & 1/2 \\ 1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$$

Goal: Recover the gradient sum by solving the decoding vector $\boldsymbol{v}$ from

$$\boldsymbol{E}\boldsymbol{v} = \mathbf{1}_K.$$

Gradient Coding with Stragglers

Stragglers: slow or non-responsive workers.



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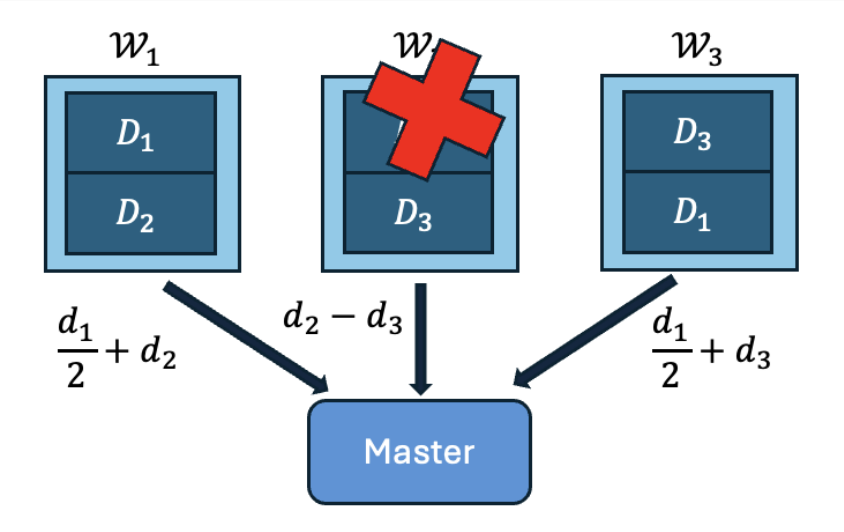

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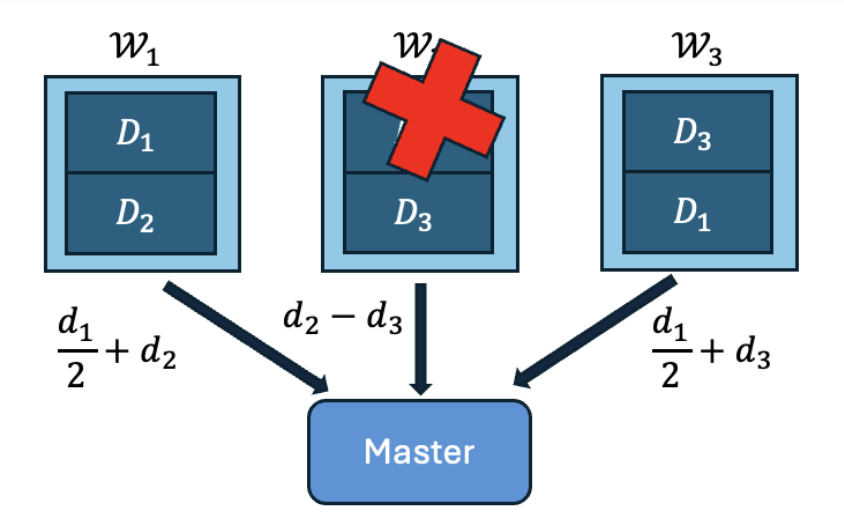

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- **Goal**: find $\boldsymbol{v}$ that minimizes

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$$\mathrm{err}(\boldsymbol{E}, S) = \frac{1}{K} \max_{\substack{\mathcal{F}\subset[N] \\ |\mathcal{F}|=N-S}} \|\boldsymbol{E}_{\mathcal{F}}\boldsymbol{v}_{\mathrm{opt}} - \mathbf{1}_K\|_2^2$$

Good Gradient Codes

Goals:

- Good straggler resilience
- Small computation workload

Good Gradient Codes

Goals:

- Small error
- Small computation workload

Good Gradient Codes

Goals:

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Good Gradient Codes

Goals:

- Small error
- Sparse encoding matrix

Existing Gradient Codes

Prior Work

- Fractional Repetition Codes [Tandon et al. (2017)]
- Bernoulli Gradient Codes [Charles et al. (2017)]
- BIBD Gradient Codes [Kadhe et al. (2019)]
- Gradient Coding from Cyclic MDS Codes and Expander Graphs [Raviv et al. (2020)]
- ... and many others

BIBD Gradient Codes

1. **Balanced Incomplete Block Design (BIBD) gradient codes [Kadhe et al. (2019)]**

$$\boldsymbol{E}^{\mathrm{B}} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \end{bmatrix}$$

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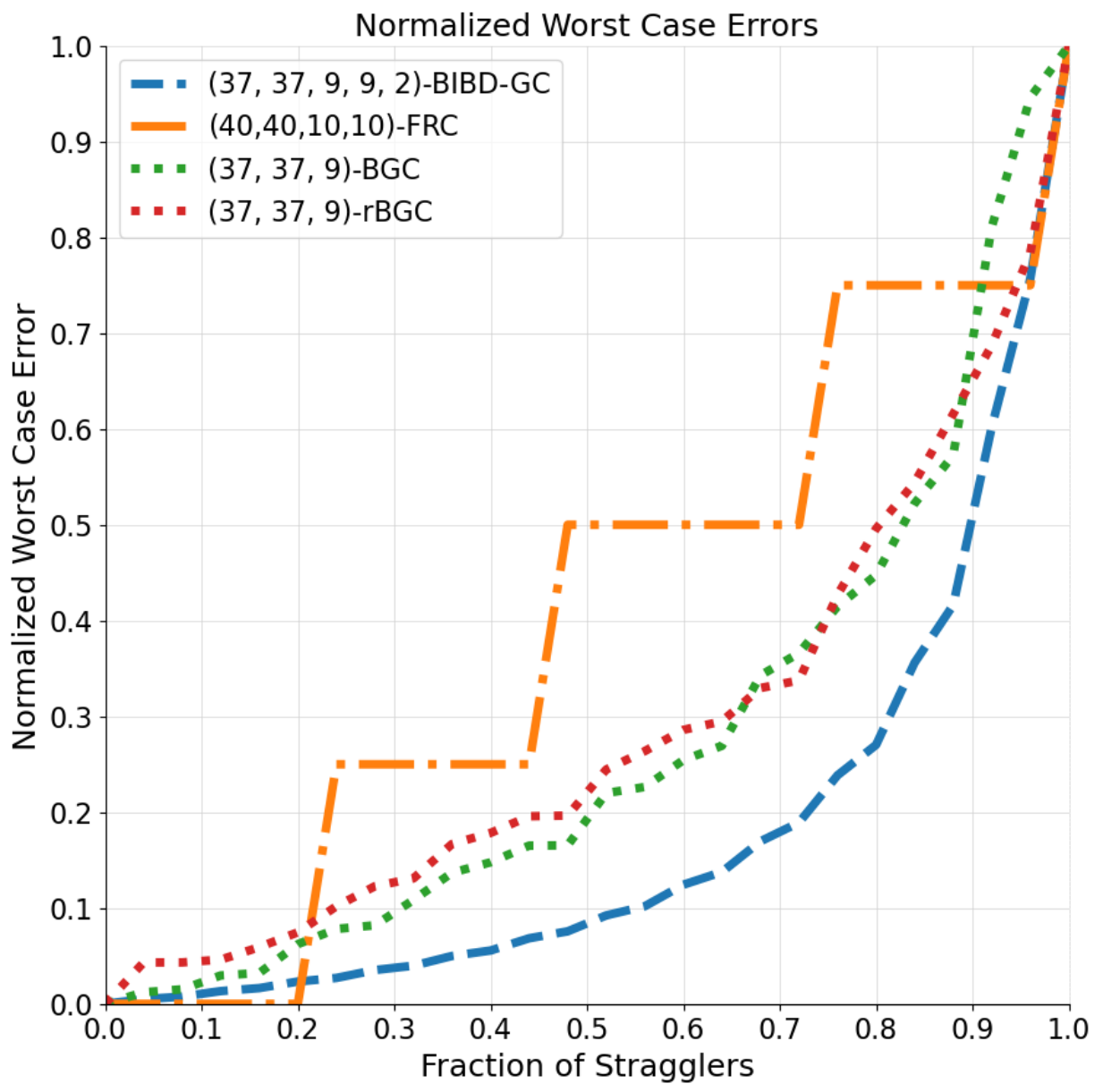
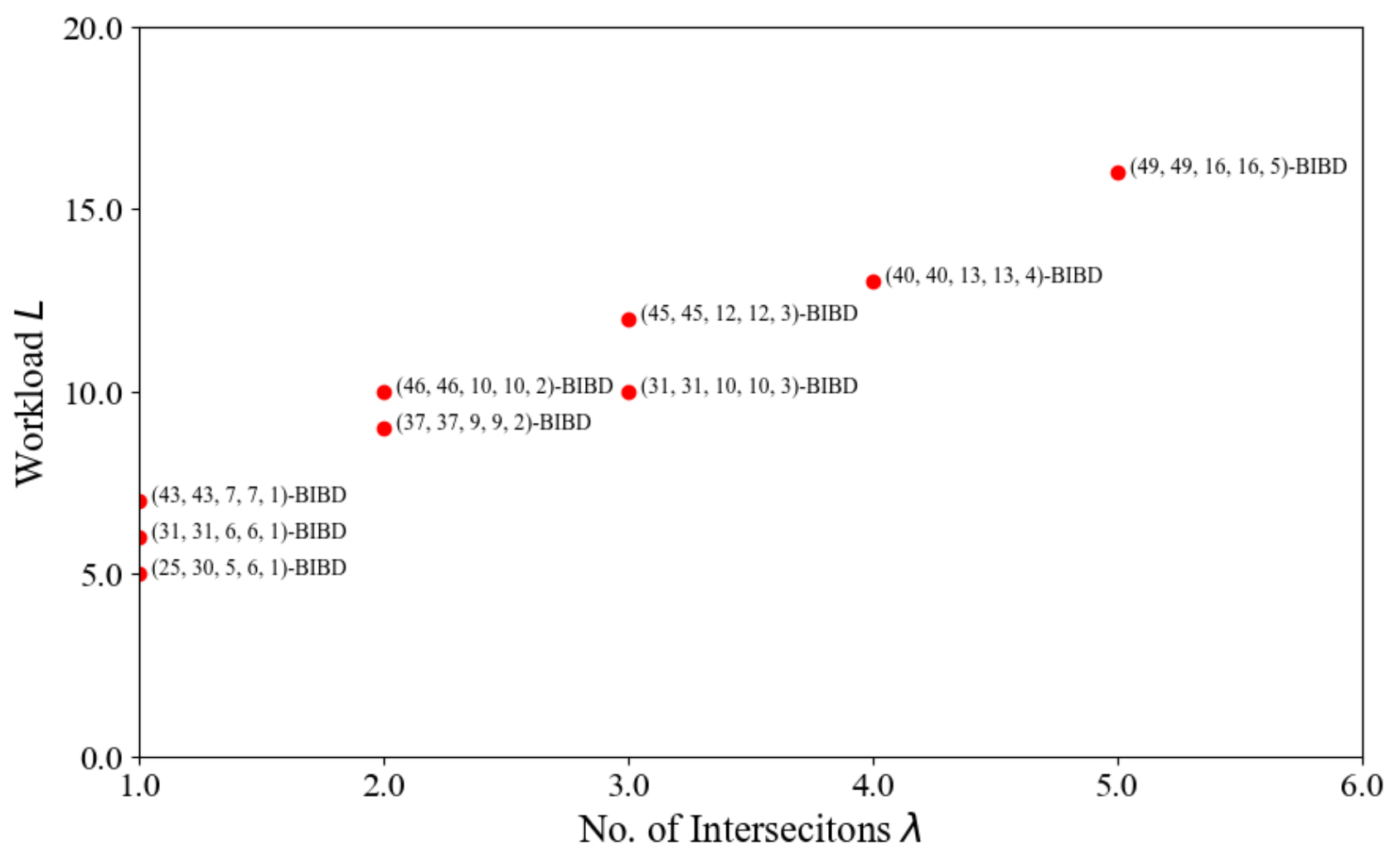
- **Pros:** the error only relates to the number of stragglers (i.e. S).

BIBD Gradient Codes



BIBD gradient codes

Cons: BIBD gradient code does NOT broadly exist.



2. **Soft BIBD Gradient Codes [Sakorikar and Wang (2022)]**

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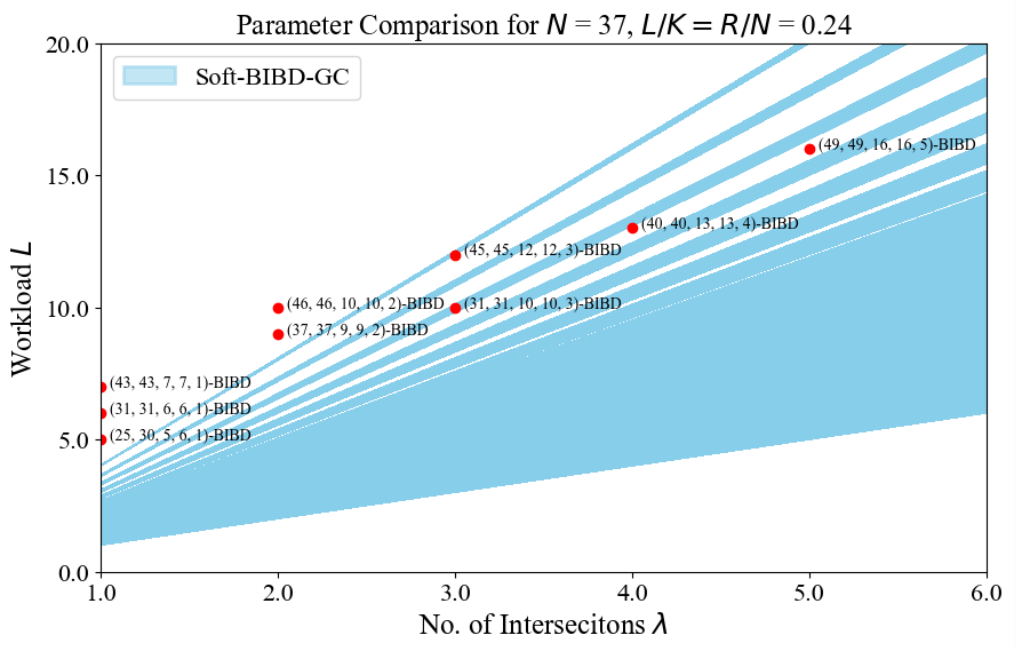
Contribution 1: Extend the range of system parameters inherent in BIBD gradient codes.



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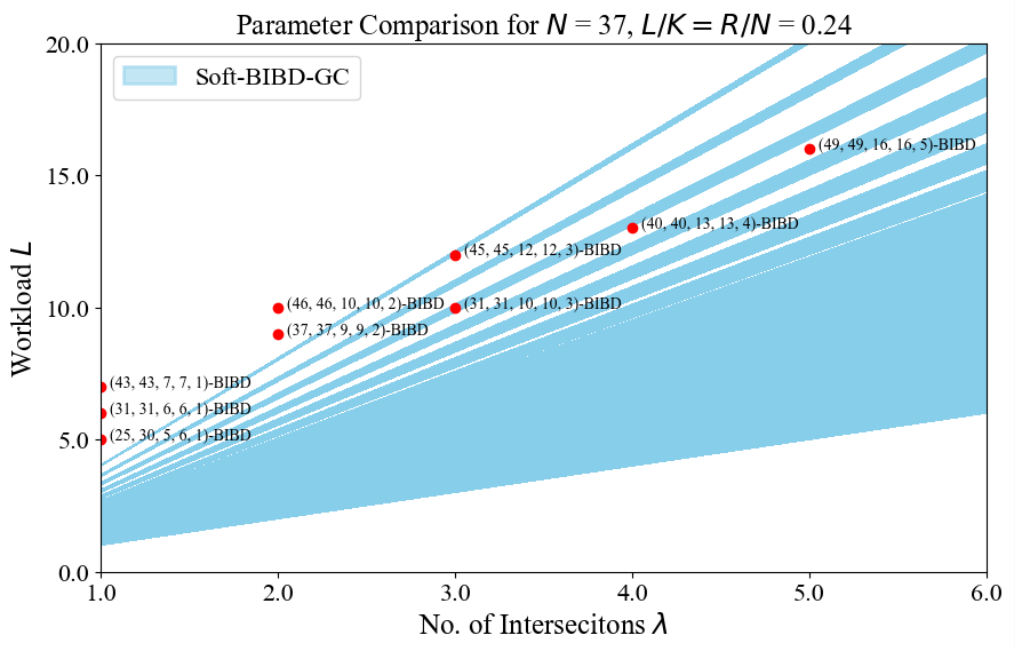

Contribution 2:

$$\mathbb{E}\left[\mathrm{err}(\boldsymbol{E}^{\mathrm{soft}}, S)\right] = \mathrm{err}(\boldsymbol{E}^{\mathrm{B}}, S).$$

Limitations of BIBD gradient codes

Limits of Soft BIBD gradient codes:

1. Complicated Construction
2. Binary Entries
3. Good Performance is not guaranteed with high probability

Motivation for a new gradient code

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Motivation for a new gradient code

1. Simple Construction for a encoding matrix
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2. Real-valued entries
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Motivation for a new gradient code

In the context of **approximate recovery**:

1. Simple Construction for a sparse encoding matrix
2. Real-valued entries
3. Good performance can be guaranteed with high probability
4. Extend the system parameter region

Sparse Gaussian Gradient Codes

SG grdient codes' Contributions

Contribution 1: Extend the range of system parameters.

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Theorem 1 (sparse Gaussian GC's parameter region)

An $(N, K, L, R, \lambda, \gamma)$ Sparse Gaussian gradient code exists if $L \leq K$ and

$$\max\left\{\frac{\lambda}{L}, \frac{(N-1)(L^2 - K\lambda) + L^2}{KL}\right\} \leq \gamma \leq 1.$$

SG Grdient Codes' Contributions

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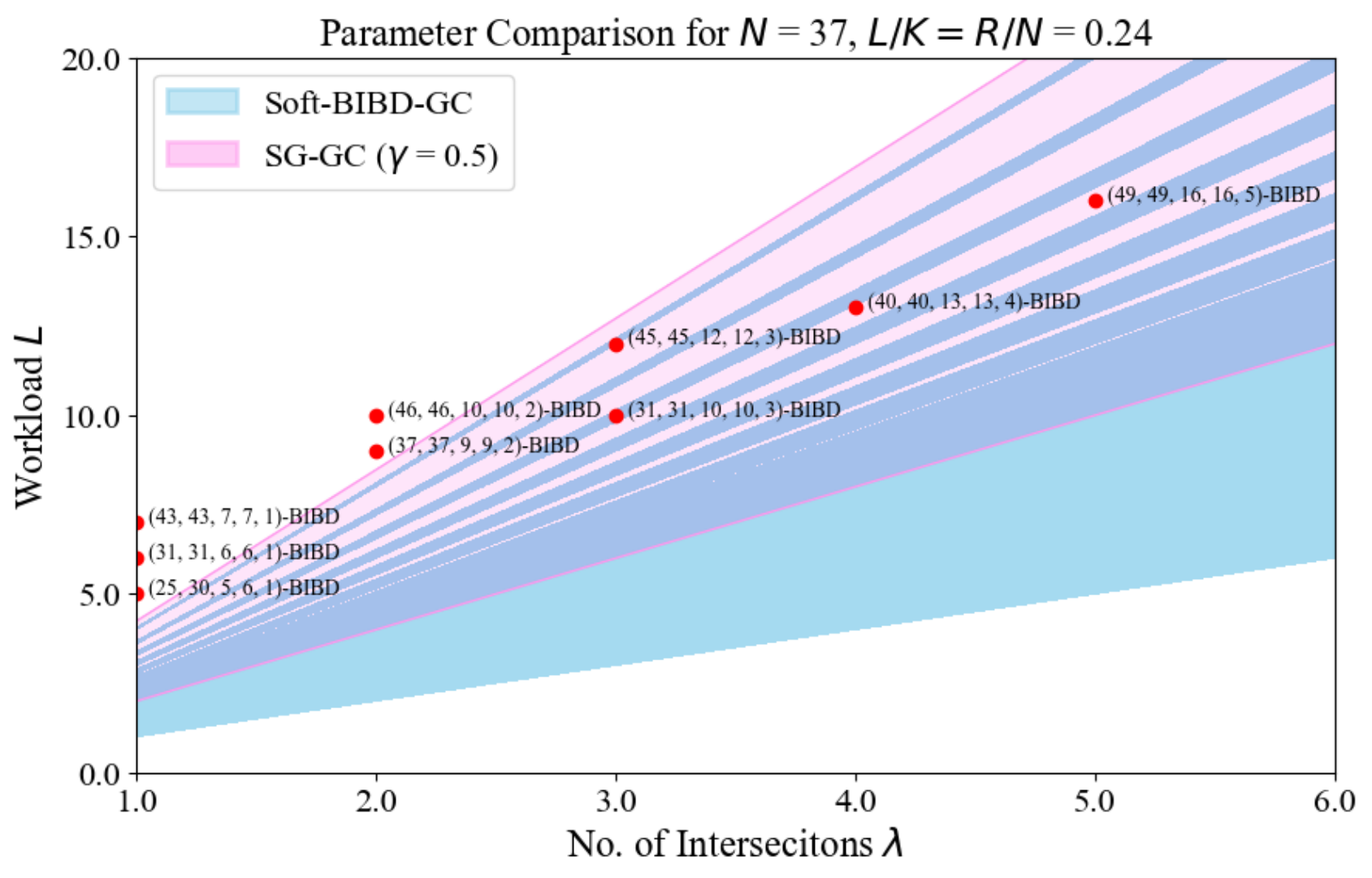

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Contribution 2: Retain a similar error performance as the BIBD gradient code with high probability.

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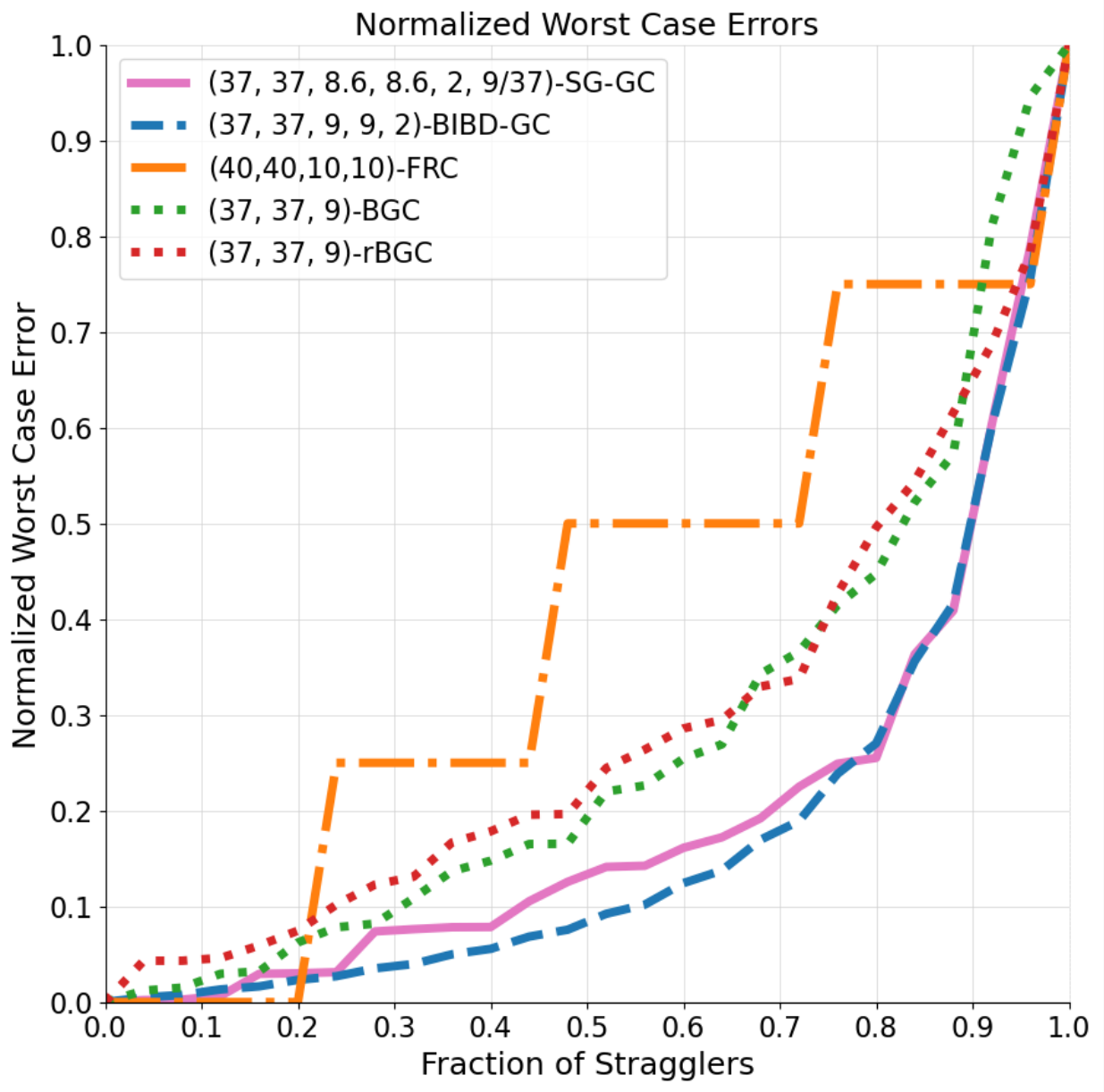
Theorem 2 (Informal) high-probability error bound

Under mild conditions: If $K = \Theta(N), L = \Theta(N), \lambda = \Theta(N), \gamma = \Theta(1)$, and $S = O(N^{\alpha})$ for some $0 < \alpha < 1$, then

$$\mathbb{P}\left[\mathrm{err}(\boldsymbol{E}^{\mathrm{SG}}, S) \leq \mathrm{err}(\boldsymbol{E}^{\mathrm{B}}, S) + o(1)\right] \geq 1 - o(1).$$

SG Grdient Codes' Contributions



SG Grdient Codes' Contributions

Contribution 3: Real-valued encoding matrix.

Contribution 4: Simple construction.

SG Gradient Codes' Construction

The encoding matrix for $(N, K, L, R, \lambda, \gamma)$ *sparse Gaussian gradient code*, denoted as $\boldsymbol{E}^{\mathrm{SG}}$, is constructed by

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$$\boldsymbol{\Sigma} = \begin{bmatrix} b & c & \cdots & c \\ c & b & \cdots & c \\ \vdots & \vdots & \ddots & \vdots \\ c & c & \cdots & b \end{bmatrix}.$$

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Each entry $\boldsymbol{B}_{ij} \overset{\text{i.i.d.}}{\sim} \mathrm{Bern}(\gamma)$.

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Remark

In this way, we ensure that a γ fraction of the encoding matrix is non-zero in expectation.

SG Gradient Codes' Construction

- Each column has the same number of non-zeros.
- Each row has the same number of non-zeros.
- Any two distinct columns incident at the same number of rows.

SG Gradient Codes' Construction

To simulate BIBD property

- $\mathbb{E}\left[\boldsymbol{E}_{ij}^{\mathrm{SG}}\right] = L/K;$
- $\mathbb{E}\left[\boldsymbol{E}_{ij}^{\mathrm{SG}}\boldsymbol{E}_{ik}^{\mathrm{SG}}\right] = \lambda/K;$
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Recall two other conditions

- $0 < \gamma < 1$;
- $\boldsymbol{\Sigma}$ is positive semi-definite.

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To simulate BIBD property

- $\mathbb{E}\left[\boldsymbol{E}_{ij}^{\mathrm{SG}}\right] = L/K$;
- $\mathbb{E}\left[\boldsymbol{E}_{ij}^{\mathrm{SG}}\boldsymbol{E}_{ik}^{\mathrm{SG}}\right] = \lambda/K$;
- $\mathbb{E}\left[\boldsymbol{E}_{ij}^{\mathrm{SG}}\boldsymbol{E}_{ij}^{\mathrm{SG}}\right] = L/K$.

Recall two other conditions

- $0 < \gamma < 1$;
- $\boldsymbol{\Sigma}$ is positive semi-definite.

Theorem 1 (sparse Gaussian GC's parameter region)

An $(N, K, L, R, \lambda, \gamma)$ Sparse Gaussian gradient code exists if $L \leq K$ and

$$\max\left\{\frac{\lambda}{L}, \frac{(N-1)(L^2 - K\lambda) + L^2}{KL}\right\} \leq \gamma \leq 1.$$

Conclusion

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Future work:

- Experiments: Run real ML tasks on clouds with SG gradient code.
- Employ other sparsifiers: Beneczur and Karger's Algorithm.

Questions?

References

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